

Deformation Theory Seminar [13]

[1]

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Applications of Bend and Break (Continuation)

- Ref • Debarre "Higher dimensional algebraic geometry"
• Kollár, Mori "Birational geometry of Algebraic geometry"
• Matsuki "Introduction to the Mori program"
• Cascini "Rational curves on complex manifolds"

<Last time>

Bend and Break I

$$\dim_{\mathbb{C}} \text{Mor}(C, X; f|_{\mathbb{C}}) \geq 1 \quad (\Leftrightarrow -K_X \cdot f_* C - g(C) \dim X \geq 1)$$

$\xRightarrow{\text{bend}} f_* C \sim f'_* C + Z$ where Z : rational 1-cycle
 break γ : ration'l curve through $f(C)$

Bend and Break II

$$\dim_{\mathbb{C}} \text{Mor}(\mathbb{P}^1, X; f|_{\mathbb{P}^1}) \geq 2 \quad (\Leftrightarrow -K_X \cdot f_* \mathbb{P}^1 - \dim X \geq 2)$$

$\Rightarrow f_* \mathbb{P}^1 \sim \sum a_i L_i$ at least two comp.
 $\rightsquigarrow$ until $\exists$ ration'l curve of antican. deg $\leq \dim X + 1$
repeat

<Application I>

Thm 1 [D] Thm 3.14 [KM] Thm 1.10

X : sm proj w/ $-K_X$: ample, $\dim X = n$.
Through any point of $x \in X$, $\exists$ rat'l curve of antican. deg $\leq n+1$

proof Let $x \in X$, In order to apply B-B it's enough to produce a curve $f: C \rightarrow X$ s.t. $-K_X \cdot f_* C - n g(C) \geq 1$

But unfortunately, there is no known way to achieve that except in positive char. (see also last my note)

For k w/ $\text{char } k = p > 0 \Rightarrow \exists \text{ Frobenius!}$

$F: C_{(1)} \rightarrow C$ where the abstract scheme $C_{(1)}$ is the same as C of $\deg = p$. but its k -str is defined as

$$C \rightarrow \text{Spec } k \rightarrow \text{Spec } k$$

$$x \mapsto x^p.$$

$\rightsquigarrow$ Iterating F , we get $F_m: C_{(m)} \rightarrow C$ of $\deg = p^m$

$$\text{Now } -K_X \cdot (f \circ F_m)_* C_m - n g(C_m)$$

$$= -p^m K_X \cdot f_* C_m - n g(C_m) \gg 0 \quad m \gg 0$$

Then by B-B, $\exists$ rat'l curve E w/ $-K_X \cdot E \leq n+1$ $\nexists$ positive char.

Q How to pass from our original situation to one over a field of char $p > 0$?

From here, k w/ $\text{char } k = 0$.

Strategy (Reduction mod p)

<Spread out>

Embed $X \subset \mathbb{P}^n$, X is defined by a fin set of equations.

Let $R := \mathbb{Z}[a_{ij}]$: finite type $\mathbb{Z}$ -alg gen. by coeff of these eq's and coordinates of x .

We obtain "model" $\tilde{X} \rightarrow \text{Spec } R$ w/ x_R

$$\begin{array}{ccccc} \tilde{X} & \longleftrightarrow & \tilde{X}_\eta & \longleftarrow & X \\ \downarrow & & \downarrow & & \downarrow \\ \text{Spec } R & \longleftrightarrow & \text{Spec } \text{Frac}(R) & \longleftarrow & \text{Spec } \text{Frac}(R) = k \\ & & \text{generic fiber} & & \text{base change} \end{array}$$

<Fact>

- Eliminating finitely many primes (appearing in denominator)
 $\exists U \subset \text{Spec } R$ open dense s.t. $\mathcal{X}$ is smooth of dim n / U .
- Ampleness is open property in family
 (see Lazarsfeld "Positivity in Algebraic Geometry I", Thm 1.2.17)
 $\Rightarrow$ relative dualizing sheaf $\omega_{\mathcal{X}/U}$ is ample on all fibers
 $\Rightarrow \forall m$: maximal ideal of R in U
 fiber on m is Fano of dim n .

<Observation>

- $\text{Spec } R = \text{Spec } \mathbb{Z}[x]/(x^2+1)$
 $\begin{cases} p=3 \Rightarrow x^2+1=0 \text{ no solution in } \mathbb{F}_3 \text{ then } R/(3, x^2+1) \cong \mathbb{F}_9 \\ p=5 \Rightarrow x^2+1=0 \text{ then } x=\pm 2 \text{ in } \mathbb{F}_5 \text{ then } R/(5, x-2) \cong \mathbb{F}_5 \end{cases}$
 $\Rightarrow$ For each maximal ideal m of R , R/m is finite.

- Intersection of all maximal ideals of R is $\{0\}$

$\Rightarrow$ Maximal ideals are dense in $\text{Spec } R$.

<Conclusion>

Now $\exists$ quasi-proj scheme

key! "universal" bound

$$P: \text{Mor}_{\leq n+1}(P', \mathcal{X}, 0 \mapsto \mathcal{X}_R) \rightarrow \text{Spec } R.$$

By BB, fiber over a closed point of $U \subset_{\text{dense}} \text{Spec } R$ is nonempty.
 $\Rightarrow$ generic fiber is also nonempty!

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Thm 2 [D] Thm 3.6 [KM] Thm 1.13 [M] Thm 10-1-1

X : sm proj var.

H : ample

Assume $\exists C'$: irred curve s.t. $-K_X \cdot C' > 0$.

$\Rightarrow \exists$ rational curve $E \subset X$ s.t.

$$\left\{ \begin{array}{l} \bullet 0 < -K_X \cdot E \leq \dim X + 1 \\ \bullet \frac{-K_X \cdot E}{H \cdot E} \geq \frac{-K_X \cdot C'}{H \cdot C'} \end{array} \right.$$

Remark! • [D], [M] states for more general case (X is sm along C') i.e. X may be singular.

$\left(\begin{array}{l} H \cdot E \leq 2 \dim X \frac{H \cdot C'}{-K_X \cdot C'} \text{ \& } E \text{ through given pt.} \\ \bullet \text{ [KM] say nothing about the position of rat'l curve.} \end{array} \right)$

Idea of proof

Consider ratio $\lambda = \frac{-K_X \cdot C'}{H \cdot C'} \Leftrightarrow (\lambda H + K_X) \cdot C' = 0$.

Using B-B, $C' \sim C'' + Z$ Z : rational 1-cycle.

$$\lambda = \frac{-K_X \cdot (C'' + Z)}{H \cdot (C'' + Z)} \leq \max \left\{ \frac{-K_X \cdot C''}{H \cdot C''}, \frac{-K_X \cdot Z}{H \cdot Z} \right\}$$

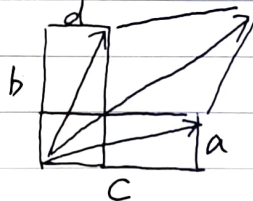
$$Z = \sum E_i$$

$\rightsquigarrow$ repeat $\lambda \leq \frac{-K_X \cdot E_i}{H \cdot E_i}$

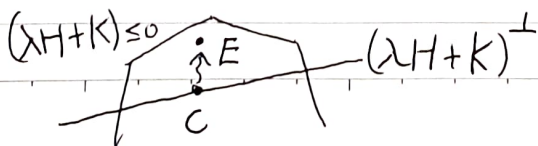
w/ $-K_X \cdot E_i \leq \dim X + 1$

we use

$$\frac{a+b}{c+d} \leq \max \left\{ \frac{a}{c}, \frac{b}{d} \right\}$$



[Idea!] $\lambda \leq \frac{-K_X \cdot E}{H \cdot E} \Rightarrow (\lambda H + K_X) \cdot E \leq 0$



Main application $\Downarrow$

Thm (Cone theorem for sm case) [D] Thm 6.1 [KM] Thm 1.24 [M] Thm 10-2-1

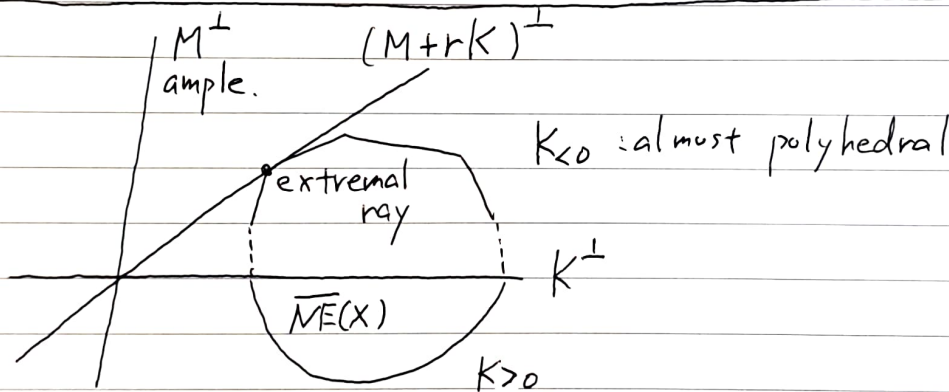
X : non-singular proj var.

- (1) $\exists$ countably many rational curves $C_i \subset X$
s.t. $0 < -K_X \cdot C_i \leq \dim X + 1$.

$$\overline{NE}(X) = \overline{NE}(X)_{K \geq 0} + \sum_i \mathbb{R}_{\geq 0} [C_i]$$

- (2) For any $\varepsilon > 0$ and H : ample

$$\overline{NE}(X) = \overline{NE}(X)_{(K+\varepsilon H) \geq 0} + \sum_{\text{fin}} \mathbb{R}_{\geq 0} [C_i]$$



proof skip, see text books!

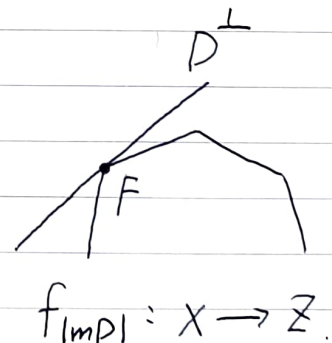
Def (Contraction thm)

F : extremal face $\exists \text{cont}_F: X \rightarrow Z$

- (1) C : irred $\subset X$

$$\text{cont}_F(C) = \{pt\} \iff [C] \in F$$

- (2) $(\text{cont}_F)_* \mathcal{O}_X = \mathcal{O}_Z$



- (2) suggests $\text{cont}_F =$ its own stein factorization.

$$\text{cont}_F \begin{cases} \cdot \text{fiber type} & \dim X > \dim Z \\ \cdot \text{divisorial} & F_X(f) = \text{div}, \text{cont}_F: \text{bir.} \\ \cdot \text{small} & F_X(f) = \text{codim} \geq 2 \leadsto \text{create new sing. (we need flip!)} \end{cases}$$

(Example). S : sm surf.

For surface, assume $R = \mathbb{R}_{>0}[C]$ $K_S \cdot R < 0$.

$\text{contr} R : S \rightarrow Z$ exists and is one of the following.

- $$\begin{cases} (1) Z: \text{sm surf} & S = \text{Bl}_{pt} Z \quad w/ \quad P(Z) = P(S) - 1 \\ (2) Z: \text{sm curv} & S: \text{minimal ruled surf}/Z \quad w/ \quad P(S) = 2. \\ (3) Z: pt, & S \cong \mathbb{P}^2. \end{cases}$$

(1) $\Rightarrow$ divisorial

(2), (3) $\Rightarrow$ Mori fiber space

Note that $C \cdot (K_S + C) = \deg w = -2$

$$\Rightarrow C^2 = -2 - K_S \cdot C \quad w/ \quad 0 < -K_S \cdot C \leq 2 + 1 = 3$$

$$\Rightarrow C^2 = -1, 0, 1$$

$$\begin{array}{ccc} \uparrow & \uparrow & \uparrow \\ (1) & (2) & (3) \end{array}$$

For (3), $C^2 > 1$ this is in the interior of $NE(S)$.

but we assume C is extremal

$\Rightarrow P(S) = 1$, C : ample

Now a map contracting ample div contracts var to pt

so $Z = pt$, and by some discussion $S \cong \mathbb{P}^2$.

For 3-fold, See [KM] Thm 1.32.

Another Application of B-B $\Downarrow$

<Hartshorne conjecture>

Consider $X = \mathbb{P}^n \Rightarrow -K_X = (n+1)H$ H : hyperplane section.
 we know that $\mathbb{P}^n$ is covered by lines L w/ $H \cdot L = 1$
 $\Rightarrow -K_X \cdot L = n+1$ "maximal" degree!

Thm (See [C] or Kollár "Rational curves on algebraic varieties"
 Mori "Projective manifolds with ample tangent bundle")
 X : Proj Fano
 smallest degree of a covering family of rational curves
 is $n+1 \Rightarrow X \cong \mathbb{P}^n$.

Cor (Mori-Hartshorne)
 T_X : ample $\Rightarrow X \cong \mathbb{P}^n$

By B-B, $\exists f: \mathbb{P}^1 \rightarrow X$ w/ $0 < -K_X \cdot f(\mathbb{P}^1) \leq n+1$.

Since T_X is ample, f^*T_X is the direct sum of ample line bundle
 $\Rightarrow f^*T_X = \bigoplus_{i=1}^n \mathcal{O}_{\mathbb{P}^1}(a_i)$ for some $a_i > 0$ w/ $\sum a_i \leq n+1$.

Since $T_{\mathbb{P}^1} \cong \mathcal{O}(2) \rightarrow f^*T_X$ is nonzero, $a_1 = 2$.

$\Rightarrow a_1 = 2, a_2, \dots, a_n = 1$

$\Rightarrow \deg = n+1$! then we apply thm. $\#$

See Mori's original paper
 Cor 7, Thm 8.